

MH1201 Linear Algebra II

Tutorial 10 (Week 12 & 13) – Problems & Solutions, Advanced Methods Edition

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Overview

This tutorial emphasizes:

- Gram–Schmidt in abstract inner product spaces and stability on orthogonal inputs;
- producing orthonormal bases that preserve a T -invariant flag, hence preserving upper-triangularity;
- orthogonal polynomials on $[-1, 1]$ and Fourier coefficients in an orthonormal basis;
- inner products defined by Hermitian positive definite matrices and complex Gram–Schmidt;
- orthogonal complements: $W \subseteq W^{\perp\perp}$ always, and $W = W^{\perp\perp}$ in finite dimensions;
- explicit orthogonal decomposition $x = a + b$ with $a \in V$, $b \in V^\perp$ in $\mathbb{R}^4$;
- Frobenius inner product on $\mathbb{R}^{n,n}$ and describing $D^\perp$ for diagonal D .

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Problem 1 (Gram–Schmidt on an orthogonal basis)

Problem

Let $(V, \langle \cdot, \cdot \rangle)$ be a finite-dimensional inner product space, with $\dim(V) = n \geq 1$. Let $\beta = (b_1, \dots, b_n)$ be an orthogonal ordered basis for $(V, \langle \cdot, \cdot \rangle)$. Show that applying Gram–Schmidt to β with respect to $(V, \langle \cdot, \cdot \rangle)$ yields β back.

Solution

Method 1: Projections vanish because of orthogonality

Recall the (non-normalized) Gram–Schmidt construction:

$$u_1 = b_1, \quad u_k := b_k - \sum_{j=1}^{k-1} \text{proj}_{u_j}(b_k), \quad \text{proj}_{u_j}(b_k) := \frac{\langle b_k, u_j \rangle}{\langle u_j, u_j \rangle} u_j, \quad 2 \leq k \leq n.$$

We prove by induction that $u_k = b_k$ for all k .

For $k = 1$, $u_1 = b_1$ is immediate.

Assume $u_j = b_j$ for all $j < k$. Since β is orthogonal, for each $j < k$,

$$\langle b_k, u_j \rangle = \langle b_k, b_j \rangle = 0.$$

Hence each projection term is 0, so

$$u_k = b_k - \sum_{j=1}^{k-1} 0 = b_k.$$

Thus Gram–Schmidt returns the same ordered orthogonal basis β .

If β is already orthogonal, Gram–Schmidt outputs β unchanged.

Method 2: Uniqueness from nested spans

In Gram–Schmidt, the output vectors satisfy:

$$\begin{aligned} u_k &\in \text{span}(b_1, \dots, b_k), \\ \text{span}(u_1, \dots, u_k) &= \text{span}(b_1, \dots, b_k), \\ \langle u_k, u_j \rangle &= 0 \quad (j < k). \end{aligned}$$

Because β is orthogonal, the vectors in $\text{span}(b_1, \dots, b_k)$ that are orthogonal to $\text{span}(b_1, \dots, b_{k-1})$ are precisely the scalar multiples of b_k . Gram–Schmidt subtracts only a vector from the preceding span, so the b_k -component remains 1. Hence $u_k = b_k$.

Method 3: Orthogonal projection onto the previous flag

For each k , define the previous flag subspace

$$F_{k-1} := \text{span}(b_1, \dots, b_{k-1}), \quad F_0 := \{0\}.$$

The Gram–Schmidt step may be written as

$$u_k = b_k - \text{proj}_{F_{k-1}}(b_k),$$

where $\text{proj}_{F_{k-1}}(b_k)$ denotes the orthogonal projection of b_k onto F_{k-1} . Since $(b_1, \dots, b_n)$ is orthogonal, for each $j < k$,

$$\langle b_j, b_k \rangle = 0.$$

Therefore b_k is orthogonal to every vector in F_{k-1} , so

$$\text{proj}_{F_{k-1}}(b_k) = 0.$$

Hence

$$u_k = b_k.$$

Thus every Gram–Schmidt step leaves the input vector unchanged.

$$\boxed{u_k = b_k \text{ for every } k = 1, \dots, n.}$$

Convention note. This conclusion is for the non-normalized Gram–Schmidt algorithm. If one’s definition of Gram–Schmidt includes normalization at every step, then the output is instead

$$\left(\frac{b_1}{\|b_1\|}, \dots, \frac{b_n}{\|b_n\|} \right).$$

Thus the precise statement depends on whether “Gram–Schmidt” means orthogonalisation or orthonormalisation.

Problem 2 (Preserving upper-triangularity)

Problem

Let

$$\beta := ((1, 0, 0), (1, 1, 1), (1, 1, 2)),$$

which is an ordered basis for $\mathbb{R}^3$ (with the standard inner product $\cdot$). Let $T \in \mathcal{L}(\mathbb{R}^3)$, and suppose that $[T]_{\beta}^{\beta} \in \mathbb{R}^{3,3}$ is upper triangular. Derive an orthonormal ordered basis γ for $(\mathbb{R}^3, \cdot)$ for which $[T]_{\gamma}^{\gamma} \in \mathbb{R}^{3,3}$ is upper triangular.

Solution

Method 1: Gram–Schmidt preserves the flag; hence upper triangular

Write $\beta = (b_1, b_2, b_3)$ with

$$b_1 = (1, 0, 0), \quad b_2 = (1, 1, 1), \quad b_3 = (1, 1, 2).$$

Since $[T]_{\beta}^{\beta}$ is upper triangular, we have the equivalent T -invariance of the coordinate flag:

$$\begin{aligned} T(\text{span}(b_1)) &\subseteq \text{span}(b_1), \\ T(\text{span}(b_1, b_2)) &\subseteq \text{span}(b_1, b_2), \\ T(\text{span}(b_1, b_2, b_3)) &\subseteq \text{span}(b_1, b_2, b_3) = \mathbb{R}^3. \end{aligned}$$

Apply Gram–Schmidt to β in the standard inner product.

Step 1: Compute γ by Gram–Schmidt. Let

$$u_1 = b_1, \quad u_2 = b_2 - \text{proj}_{u_1}(b_2), \quad u_3 = b_3 - \text{proj}_{u_1}(b_3) - \text{proj}_{u_2}(b_3).$$

Compute:

$$\text{proj}_{u_1}(b_2) = \frac{b_2 \cdot u_1}{u_1 \cdot u_1} u_1 = \frac{1}{1} b_1 = b_1, \quad \Rightarrow \quad u_2 = b_2 - b_1 = (0, 1, 1).$$

Similarly,

$$\text{proj}_{u_1}(b_3) = \frac{b_3 \cdot u_1}{u_1 \cdot u_1} u_1 = b_1, \quad \Rightarrow \quad b_3 - \text{proj}_{u_1}(b_3) = (0, 1, 2).$$

Now

$$\text{proj}_{u_2}(b_3) = \text{proj}_{u_2}((0, 1, 2)) = \frac{(0, 1, 2) \cdot (0, 1, 1)}{(0, 1, 1) \cdot (0, 1, 1)} (0, 1, 1) = \frac{3}{2} (0, 1, 1).$$

Thus

$$u_3 = (0, 1, 2) - \frac{3}{2} (0, 1, 1) = \left(0, -\frac{1}{2}, \frac{1}{2}\right).$$

Normalize:

$$\|u_1\| = 1, \quad \|u_2\| = \sqrt{2}, \quad \|u_3\| = \sqrt{\frac{1}{2}}.$$

Hence an orthonormal basis is

$$\gamma = (g_1, g_2, g_3) := \left((1, 0, 0), \left(0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right), \left(0, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right) \right).$$

$$\gamma = \left((1, 0, 0), \left(0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right), \left(0, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right) \right) \text{ is orthonormal.}$$

Step 2: Show $[T]_\gamma^\gamma$ is upper triangular. A key Gram–Schmidt property is

$$g_k \in \text{span}(b_1, \dots, b_k) \quad (k = 1, 2, 3).$$

Therefore

$$\text{span}(g_1, \dots, g_k) = \text{span}(b_1, \dots, b_k) \quad (k = 1, 2, 3).$$

Since $T(\text{span}(b_1, \dots, b_k)) \subseteq \text{span}(b_1, \dots, b_k)$, we get

$$T(\text{span}(g_1, \dots, g_k)) \subseteq \text{span}(g_1, \dots, g_k) \quad (k = 1, 2, 3).$$

This is exactly the condition that the matrix of T in the ordered basis γ is upper triangular.

$$[T]_\gamma^\gamma \text{ is upper triangular.}$$

Method 2: Upper-triangular similarity via an upper-triangular change-of-basis matrix

Let $B = [T]_\beta^\beta$ (upper triangular). Let P be the change-of-basis matrix from γ -coordinates to β -coordinates:

$$P := [\text{id}]_\beta^\gamma.$$

Because Gram–Schmidt produces $g_k \in \text{span}(b_1, \dots, b_k)$, the coordinate vector $[g_k]_\beta$ has no components in $b_{k+1}, \dots, b_3$, so P is upper triangular with nonzero diagonal (hence invertible). Then

$$[T]_\gamma^\gamma = P^{-1} [T]_\beta^\beta P = P^{-1} B P.$$

A product (and inverse) of upper-triangular invertible matrices is upper triangular, hence $[T]_\gamma^\gamma$ is upper triangular.

Method 3: Flag-slice construction without full Gram–Schmidt

Define the flag associated with the original ordered basis:

$$F_k := \text{span}(b_1, \dots, b_k), \quad k = 1, 2, 3.$$

Because $[T]_\beta^\beta$ is upper triangular, the flag is T -invariant:

$$T(F_k) \subseteq F_k \quad (k = 1, 2, 3).$$

Instead of carrying out Gram–Schmidt mechanically, construct the orthonormal basis by taking the new unit direction in each orthogonal slice:

$$g_k \in F_k \cap F_{k-1}^\perp, \quad \|g_k\| = 1, \quad F_0 := \{0\}.$$

For this problem,

$$F_1 = \text{span}\{(1, 0, 0)\},$$

so

$$g_1 = (1, 0, 0).$$

Next,

$$F_2 = \text{span}\{(1, 0, 0), (1, 1, 1)\} = \{(a, b, b) : a, b \in \mathbb{R}\}.$$

The vector in F_2 perpendicular to g_1 must have first coordinate 0, hence is proportional to $(0, 1, 1)$. Therefore

$$g_2 = \frac{1}{\sqrt{2}}(0, 1, 1).$$

Finally, since $F_3 = \mathbb{R}^3$, the last direction is the unit vector perpendicular to F_2 . A vector perpendicular to both $(1, 0, 0)$ and $(0, 1, 1)$ is proportional to $(0, -1, 1)$, so we may take

$$g_3 = \frac{1}{\sqrt{2}}(0, -1, 1).$$

Thus

$$\gamma = \left((1, 0, 0), \frac{1}{\sqrt{2}}(0, 1, 1), \frac{1}{\sqrt{2}}(0, -1, 1) \right).$$

Moreover,

$$\text{span}(g_1, \dots, g_k) = F_k \quad (k = 1, 2, 3).$$

Since each F_k is T -invariant, the ordered basis γ also gives an invariant flag. Hence

$$[T]_\gamma^\gamma \text{ is upper triangular.}$$

This method is usually faster than computing all projection coefficients.

Method 4: QR-factorisation and triangular similarity

Let

$$C := [b_1 \ b_2 \ b_3] = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix}.$$

Applying Gram–Schmidt to the columns of C gives a QR factorisation

$$C = QR,$$

where the columns of Q form the orthonormal basis γ and R is invertible upper triangular.

Let

$$B := [T]_{\beta}^{\beta}.$$

By hypothesis, B is upper triangular. Since R is an upper-triangular change-of-coordinates matrix between the two flag-adapted bases, the matrix of T in the orthonormal basis is

$$[T]_{\gamma}^{\gamma} = RBR^{-1}.$$

The inverse of an invertible upper-triangular matrix is upper triangular, and products of upper-triangular matrices are upper triangular. Therefore

$[T]_{\gamma}^{\gamma}$ is upper triangular.

This method packages the whole argument into the fact that Gram–Schmidt is exactly QR factorisation.

Problem 3 (Orthogonal polynomials on $[-1, 1]$)

Problem

Define a function $\langle \cdot, \cdot \rangle : \mathcal{P}_3(\mathbb{R}) \times \mathcal{P}_3(\mathbb{R}) \rightarrow \mathbb{R}$ as follows:

$$\forall P, Q \in \mathcal{P}_3(\mathbb{R}) : \quad \langle P, Q \rangle := \int_{-1}^1 P(x) Q(x) dx.$$

- (a) Show that $\langle \cdot, \cdot \rangle$ is an inner product on $\mathcal{P}_3(\mathbb{R})$.
- (b) Apply Gram–Schmidt to $(1, X, X^2, X^3)$ with respect to $(\mathcal{P}_3(\mathbb{R}), \langle \cdot, \cdot \rangle)$.
- (c) Normalize your answer to Part (b) to obtain an orthonormal basis β for $(\mathcal{P}_3(\mathbb{R}), \langle \cdot, \cdot \rangle)$.
- (d) Compute the β -Fourier coefficients of $1 - X + X^2 - X^3$ with respect to $(\mathcal{P}_3(\mathbb{R}), \langle \cdot, \cdot \rangle)$.

Solution

Method 1: Direct axiom verification + explicit Gram–Schmidt computations

- (a) **Inner product axioms.** Bilinearity follows from linearity of the integral:

$$\langle aP_1 + bP_2, Q \rangle = a \langle P_1, Q \rangle + b \langle P_2, Q \rangle, \quad \langle P, aQ_1 + bQ_2 \rangle = a \langle P, Q_1 \rangle + b \langle P, Q_2 \rangle.$$

Symmetry holds because $P(x)Q(x) = Q(x)P(x)$:

$$\langle P, Q \rangle = \int_{-1}^1 PQ = \int_{-1}^1 QP = \langle Q, P \rangle.$$

Positive definiteness: for $P \neq 0$,

$$\langle P, P \rangle = \int_{-1}^1 P(x)^2 dx > 0.$$

Indeed, $P(x)^2 \geq 0$ and is continuous; if the integral were 0, then $P(x)^2 = 0$ for all $x \in [-1, 1]$, so P vanishes on an interval, hence P is the zero polynomial, contradiction. Therefore $\langle \cdot, \cdot \rangle$ is an inner product.

$\langle \cdot, \cdot \rangle \text{ is an inner product on } \mathcal{P}_3(\mathbb{R}).$

- (b) **Gram–Schmidt on $(1, X, X^2, X^3)$.** Let $v_0 = 1, v_1 = X, v_2 = X^2, v_3 = X^3$. Define

$$\begin{aligned} u_0 &= v_0, & u_1 &= v_1 - \text{proj}_{u_0}(v_1), \\ u_2 &= v_2 - \text{proj}_{u_0}(v_2) - \text{proj}_{u_1}(v_2), & u_3 &= v_3 - \sum_{j=0}^2 \text{proj}_{u_j}(v_3). \end{aligned}$$

Compute required inner products using parity:

$$\int_{-1}^1 x \, dx = 0, \quad \int_{-1}^1 x^3 \, dx = 0, \quad \int_{-1}^1 x^5 \, dx = 0.$$

Also

$$\int_{-1}^1 1 \, dx = 2, \quad \int_{-1}^1 x^2 \, dx = \frac{2}{3}, \quad \int_{-1}^1 x^4 \, dx = \frac{2}{5}, \quad \int_{-1}^1 x^6 \, dx = \frac{2}{7}.$$

Step 1: $u_0 = 1$.

Step 2: u_1 . Since $\langle X, 1 \rangle = \int_{-1}^1 x \, dx = 0$, we get

$$u_1 = X.$$

Step 3: u_2 . Compute

$$\text{proj}_{u_0}(X^2) = \frac{\langle X^2, 1 \rangle}{\langle 1, 1 \rangle} 1 = \frac{\frac{2}{3}}{2} \cdot 1 = \frac{1}{3}.$$

Also $\langle X^2, X \rangle = \int_{-1}^1 x^3 \, dx = 0$, so $\text{proj}_{u_1}(X^2) = 0$. Hence

$$u_2 = X^2 - \frac{1}{3}.$$

Step 4: u_3 . We have $\langle X^3, 1 \rangle = \int_{-1}^1 x^3 \, dx = 0$, and

$$\text{proj}_{u_1}(X^3) = \frac{\langle X^3, X \rangle}{\langle X, X \rangle} X = \frac{\int_{-1}^1 x^4 \, dx}{\int_{-1}^1 x^2 \, dx} X = \frac{\frac{2}{5}}{\frac{2}{3}} X = \frac{3}{5} X.$$

Finally,

$$\langle X^3, u_2 \rangle = \int_{-1}^1 x^3 \left(x^2 - \frac{1}{3} \right) dx = \int_{-1}^1 \left(x^5 - \frac{1}{3} x^3 \right) dx = 0,$$

so $\text{proj}_{u_2}(X^3) = 0$. Therefore

$$u_3 = X^3 - \frac{3}{5} X.$$

Thus Gram–Schmidt yields the orthogonal list

$$\boxed{\left(1, \, X, \, X^2 - \frac{1}{3}, \, X^3 - \frac{3}{5} X \right)}.$$

(c) **Normalize to an orthonormal basis β .** Compute norms:

$$\underline{u_0 = 1}: \|u_0\|^2 = \int_{-1}^1 1 \, dx = 2, \text{ so } \|u_0\| = \sqrt{2}.$$

$$\underline{u_1 = X}: \|u_1\|^2 = \int_{-1}^1 x^2 \, dx = \frac{2}{3}, \text{ so } \|u_1\| = \sqrt{\frac{2}{3}}.$$

$$\underline{u_2 = X^2 - \frac{1}{3}}:$$

$$\|u_2\|^2 = \int_{-1}^1 \left(x^2 - \frac{1}{3}\right)^2 dx = \int_{-1}^1 \left(x^4 - \frac{2}{3}x^2 + \frac{1}{9}\right) dx = \frac{2}{5} - \frac{2}{3} \cdot \frac{2}{3} + \frac{1}{9} \cdot 2 = \frac{8}{45}.$$

$$\text{So } \|u_2\| = \sqrt{\frac{8}{45}}.$$

$$\underline{u_3 = X^3 - \frac{3}{5}X}:$$

$$\|u_3\|^2 = \int_{-1}^1 \left(x^3 - \frac{3}{5}x\right)^2 dx = \int_{-1}^1 \left(x^6 - \frac{6}{5}x^4 + \frac{9}{25}x^2\right) dx = \frac{2}{7} - \frac{6}{5} \cdot \frac{2}{5} + \frac{9}{25} \cdot \frac{2}{3} = \frac{8}{175}.$$

$$\text{So } \|u_3\| = \sqrt{\frac{8}{175}}.$$

Define the orthonormal basis $\beta = (\beta_0, \beta_1, \beta_2, \beta_3)$ by $\beta_j = u_j / \|u_j\|$:

$$\beta_0 = \frac{1}{\sqrt{2}}, \quad \beta_1 = \sqrt{\frac{3}{2}}X, \quad \beta_2 = \sqrt{\frac{45}{8}}\left(X^2 - \frac{1}{3}\right), \quad \beta_3 = \sqrt{\frac{175}{8}}\left(X^3 - \frac{3}{5}X\right).$$

$$\beta = \left(\frac{1}{\sqrt{2}}, \sqrt{\frac{3}{2}}X, \sqrt{\frac{45}{8}}\left(X^2 - \frac{1}{3}\right), \sqrt{\frac{175}{8}}\left(X^3 - \frac{3}{5}X\right)\right) \text{ is orthonormal.}$$

- (d) **β -Fourier coefficients of $f(X) = 1 - X + X^2 - X^3$.** Since β is orthonormal, the Fourier coefficient of f along β_j is

$$c_j = \langle f, \beta_j \rangle.$$

c_0 :

$$c_0 = \left\langle f, \frac{1}{\sqrt{2}} \right\rangle = \frac{1}{\sqrt{2}} \int_{-1}^1 (1 - x + x^2 - x^3) dx = \frac{1}{\sqrt{2}} \left(2 + \frac{2}{3}\right) = \frac{8}{3\sqrt{2}}.$$

c_1 :

$$\begin{aligned} c_1 &= \left\langle f, \sqrt{\frac{3}{2}}x \right\rangle = \sqrt{\frac{3}{2}} \int_{-1}^1 (1 - x + x^2 - x^3)x dx \\ &= \sqrt{\frac{3}{2}} \int_{-1}^1 (x - x^2 + x^3 - x^4) dx \\ &= \sqrt{\frac{3}{2}} \left(-\frac{2}{3} - \frac{2}{5}\right) = -\frac{16}{15}\sqrt{\frac{3}{2}}. \end{aligned}$$

c_2 :

$$c_2 = \left\langle f, \sqrt{\frac{45}{8}}\left(x^2 - \frac{1}{3}\right) \right\rangle = \sqrt{\frac{45}{8}} \int_{-1}^1 (1 - x + x^2 - x^3) \left(x^2 - \frac{1}{3}\right) dx.$$

Expanding and using parity (odd terms integrate to 0) gives

$$\int_{-1}^1 (1 - x + x^2 - x^3) \left(x^2 - \frac{1}{3} \right) dx = \frac{8}{45}.$$

Thus

$$c_2 = \sqrt{\frac{45}{8}} \cdot \frac{8}{45} = \frac{4\sqrt{5}}{15\sqrt{2}}.$$

c_3 :

$$c_3 = \left\langle f, \sqrt{\frac{175}{8}} \left(x^3 - \frac{3}{5}x \right) \right\rangle = \sqrt{\frac{175}{8}} \int_{-1}^1 (1 - x + x^2 - x^3) \left(x^3 - \frac{3}{5}x \right) dx.$$

Keeping only even powers after expansion yields

$$\int_{-1}^1 (1 - x + x^2 - x^3) \left(x^3 - \frac{3}{5}x \right) dx = -\frac{8}{175}.$$

Hence

$$c_3 = \sqrt{\frac{175}{8}} \cdot \left(-\frac{8}{175} \right) = -\frac{4\sqrt{7}}{35\sqrt{2}}.$$

Therefore the β -Fourier coefficients are

$$(c_0, c_1, c_2, c_3) = \left(\frac{8}{3\sqrt{2}}, -\frac{16}{15}\sqrt{\frac{3}{2}}, \frac{4\sqrt{5}}{15\sqrt{2}}, -\frac{4\sqrt{7}}{35\sqrt{2}} \right).$$

Method 2: Legendre-polynomial shortcut**(a) Inner product axioms.**

For $a, b \in \mathbb{R}$ and $P_1, P_2, Q \in \mathcal{P}_3(\mathbb{R})$,

$$\langle aP_1 + bP_2, Q \rangle = \int_{-1}^1 (aP_1(x) + bP_2(x))Q(x) dx = a \langle P_1, Q \rangle + b \langle P_2, Q \rangle.$$

Similarly, by linearity in the second argument,

$$\langle P, aQ_1 + bQ_2 \rangle = a \langle P, Q_1 \rangle + b \langle P, Q_2 \rangle.$$

Also,

$$\langle P, Q \rangle = \int_{-1}^1 P(x)Q(x) dx = \int_{-1}^1 Q(x)P(x) dx = \langle Q, P \rangle.$$

Finally,

$$\langle P, P \rangle = \int_{-1}^1 P(x)^2 dx \geq 0.$$

If $\langle P, P \rangle = 0$, then

$$\int_{-1}^1 P(x)^2 dx = 0.$$

Since $P(x)^2 \geq 0$ is continuous on $[-1, 1]$, this implies

$$P(x)^2 = 0 \quad \forall x \in [-1, 1].$$

Hence $P(x) = 0$ for all $x \in [-1, 1]$. Since a nonzero polynomial cannot vanish on an interval, we must have $P = 0$. Therefore

$$\langle P, P \rangle = 0 \iff P = 0.$$

So $\langle \cdot, \cdot \rangle$ is an inner product on $\mathcal{P}_3(\mathbb{R})$.

$\langle \cdot, \cdot \rangle \text{ is an inner product on } \mathcal{P}_3(\mathbb{R}).$

(b) Apply Gram–Schmidt using Legendre polynomials.

The standard Legendre polynomials are

$$L_0(x) = 1, \quad L_1(x) = x, \quad L_2(x) = \frac{1}{2}(3x^2 - 1), \quad L_3(x) = \frac{1}{2}(5x^3 - 3x).$$

They satisfy the orthogonality relation

$$\int_{-1}^1 L_m(x)L_n(x) dx = 0 \quad (m \neq n).$$

Therefore

$$(L_0, L_1, L_2, L_3)$$

is an orthogonal list in $\mathcal{P}_3(\mathbb{R})$.

We now connect this with Gram–Schmidt applied to

$$(1, X, X^2, X^3).$$

Gram–Schmidt produces an orthogonal polynomial of degree 0, then degree 1, then degree 2, then degree 3. Since L_n is also an orthogonal polynomial of degree n , the Gram–Schmidt output must agree with the Legendre polynomial L_n up to a nonzero scalar multiple.

Now rewrite L_2 and L_3 in monic-like form:

$$L_2(X) = \frac{1}{2}(3X^2 - 1) = \frac{3}{2}\left(X^2 - \frac{1}{3}\right),$$

so

$$X^2 - \frac{1}{3} = \frac{2}{3}L_2(X).$$

Similarly,

$$L_3(X) = \frac{1}{2}(5X^3 - 3X) = \frac{5}{2}\left(X^3 - \frac{3}{5}X\right),$$

so

$$X^3 - \frac{3}{5}X = \frac{2}{5}L_3(X).$$

Hence the Gram–Schmidt orthogonal list obtained from

$$(1, X, X^2, X^3)$$

is

$$\left(1, X, X^2 - \frac{1}{3}, X^3 - \frac{3}{5}X\right).$$

(c) Normalize the orthogonal list.

The Legendre norm formula is

$$\int_{-1}^1 L_n(x)^2 dx = \frac{2}{2n+1}.$$

Hence the normalized Legendre polynomial is

$$\phi_n(X) = \sqrt{\frac{2n+1}{2}} L_n(X).$$

For $n = 0, 1, 2, 3$, this gives

$$\begin{aligned}\phi_0(X) &= \frac{1}{\sqrt{2}}, \\ \phi_1(X) &= \sqrt{\frac{3}{2}}X,\end{aligned}$$

$$\begin{aligned}\phi_2(X) &= \sqrt{\frac{5}{2}} \cdot \frac{1}{2}(3X^2 - 1), \\ \phi_3(X) &= \sqrt{\frac{7}{2}} \cdot \frac{1}{2}(5X^3 - 3X).\end{aligned}$$

Equivalently, using

$$L_2(X) = \frac{3}{2} \left(X^2 - \frac{1}{3} \right), \quad L_3(X) = \frac{5}{2} \left(X^3 - \frac{3}{5}X \right),$$

we may write

$$\phi_2(X) = \sqrt{\frac{5}{2}} \cdot \frac{3}{2} \left(X^2 - \frac{1}{3} \right) = \sqrt{\frac{45}{8}} \left(X^2 - \frac{1}{3} \right),$$

and

$$\phi_3(X) = \sqrt{\frac{7}{2}} \cdot \frac{5}{2} \left(X^3 - \frac{3}{5}X \right) = \sqrt{\frac{175}{8}} \left(X^3 - \frac{3}{5}X \right).$$

Therefore an orthonormal basis is

$$\beta = \left(\frac{1}{\sqrt{2}}, \sqrt{\frac{3}{2}}X, \sqrt{\frac{45}{8}} \left(X^2 - \frac{1}{3} \right), \sqrt{\frac{175}{8}} \left(X^3 - \frac{3}{5}X \right) \right).$$

(d) **Compute the β -Fourier coefficients of $f(X) = 1 - X + X^2 - X^3$.**

Since the basis in Part (c) is the normalized Legendre basis, it is convenient to first express f in the unnormalized Legendre basis.

We use

$$L_0(X) = 1, \quad L_1(X) = X,$$

and from

$$L_2(X) = \frac{1}{2}(3X^2 - 1),$$

we get

$$3X^2 - 1 = 2L_2(X),$$

so

$$X^2 = \frac{1}{3} + \frac{2}{3}L_2(X).$$

Also, from

$$L_3(X) = \frac{1}{2}(5X^3 - 3X),$$

we get

$$5X^3 - 3X = 2L_3(X),$$

so

$$X^3 = \frac{3}{5}X + \frac{2}{5}L_3(X) = \frac{3}{5}L_1(X) + \frac{2}{5}L_3(X).$$

Now substitute these into

$$f(X) = 1 - X + X^2 - X^3.$$

Then

$$\begin{aligned} f(X) &= L_0(X) - L_1(X) + \left(\frac{1}{3}L_0(X) + \frac{2}{3}L_2(X)\right) - \left(\frac{3}{5}L_1(X) + \frac{2}{5}L_3(X)\right) \\ &= \frac{4}{3}L_0(X) - \frac{8}{5}L_1(X) + \frac{2}{3}L_2(X) - \frac{2}{5}L_3(X). \end{aligned}$$

The normalized Legendre basis is

$$\phi_n(X) = \sqrt{\frac{2n+1}{2}}L_n(X).$$

Therefore

$$L_n(X) = \sqrt{\frac{2}{2n+1}}\phi_n(X).$$

So if

$$f(X) = a_0L_0(X) + a_1L_1(X) + a_2L_2(X) + a_3L_3(X),$$

then the Fourier coefficient of f along ϕ_n is

$$c_n = a_n\sqrt{\frac{2}{2n+1}}.$$

Here,

$$a_0 = \frac{4}{3}, \quad a_1 = -\frac{8}{5}, \quad a_2 = \frac{2}{3}, \quad a_3 = -\frac{2}{5}.$$

Thus

$$\begin{aligned} c_0 &= \frac{4}{3}\sqrt{2} = \frac{8}{3\sqrt{2}}, \\ c_1 &= -\frac{8}{5}\sqrt{\frac{2}{3}} = -\frac{16}{15}\sqrt{\frac{3}{2}}, \\ c_2 &= \frac{2}{3}\sqrt{\frac{2}{5}} = \frac{4\sqrt{5}}{15\sqrt{2}}, \end{aligned}$$

and

$$c_3 = -\frac{2}{5}\sqrt{\frac{2}{7}} = -\frac{4\sqrt{7}}{35\sqrt{2}}.$$

Therefore the β -Fourier coefficients are

$$(c_0, c_1, c_2, c_3) = \left(\frac{8}{3\sqrt{2}}, -\frac{16}{15}\sqrt{\frac{3}{2}}, \frac{4\sqrt{5}}{15\sqrt{2}}, -\frac{4\sqrt{7}}{35\sqrt{2}} \right).$$

Method 3: Even–odd parity block shortcut

The inner product

$$\langle P, Q \rangle = \int_{-1}^1 P(x)Q(x) dx$$

has an important symmetry: odd functions integrate to zero over $[-1, 1]$. Therefore even polynomials are orthogonal to odd polynomials. Hence Gram–Schmidt separates into two independent blocks:

$$\text{span}\{1, X^2\} \quad \text{and} \quad \text{span}\{X, X^3\}.$$

For the even block,

$$u_0 = 1, \quad u_2 = X^2 - \frac{\langle 1, X^2 \rangle}{\langle 1, 1 \rangle} 1 = X^2 - \frac{1}{3}.$$

For the odd block,

$$u_1 = X, \quad u_3 = X^3 - \frac{\langle X, X^3 \rangle}{\langle X, X \rangle} X = X^3 - \frac{3}{5}X.$$

This recovers

$$\boxed{(1, X, X^2 - \frac{1}{3}, X^3 - \frac{3}{5}X)}$$

with much less computation. The method works whenever the interval and weight function are symmetric, so that even–odd orthogonality is available.

Problem 4 (Complex inner product via a Hermitian positive definite matrix)

Problem

Define a function $\langle \cdot, \cdot \rangle : \mathbb{C}^{3,1} \times \mathbb{C}^{3,1} \rightarrow \mathbb{C}$ as follows:

$$\forall u, v \in \mathbb{C}^{3,1} : \quad \langle u, v \rangle := u^\dagger \begin{bmatrix} 2 & 1+i & 0 \\ 1-i & 3 & 2 \\ 0 & 2 & 4 \end{bmatrix} v,$$

where † denotes conjugate transposition.

- Show that $\langle \cdot, \cdot \rangle$ is an inner product on $\mathbb{C}^{3,1}$.
- Apply Gram–Schmidt to the standard ordered basis for $\mathbb{C}^{3,1}$ with respect to $(\mathbb{C}^{3,1}, \langle \cdot, \cdot \rangle)$.
- Normalize your answer to Part (b) to obtain an orthonormal basis β for $(\mathbb{C}^{3,1}, \langle \cdot, \cdot \rangle)$.
- Compute the β -Fourier coefficients of $\begin{bmatrix} 1 \\ 0 \\ i \end{bmatrix}$ with respect to $(\mathbb{C}^{3,1}, \langle \cdot, \cdot \rangle)$.

Solution

Let

$$H := \begin{bmatrix} 2 & 1+i & 0 \\ 1-i & 3 & 2 \\ 0 & 2 & 4 \end{bmatrix}, \quad \langle u, v \rangle = u^\dagger H v.$$

We use the standard convention: conjugate-linear in the first argument and linear in the second.

Method 1: Hermitian positive definite $\Rightarrow$ inner product; then compute Gram–Schmidt explicitly

- Axiom 1: sesquilinearity.** For $a, b \in \mathbb{C}$ and $u_1, u_2, v \in \mathbb{C}^{3,1}$,

$$\langle au_1 + bu_2, v \rangle = (au_1 + bu_2)^\dagger H v = \bar{a} u_1^\dagger H v + \bar{b} u_2^\dagger H v = \bar{a} \langle u_1, v \rangle + \bar{b} \langle u_2, v \rangle,$$

and similarly $\langle u, av_1 + bv_2 \rangle = a \langle u, v_1 \rangle + b \langle u, v_2 \rangle$.

Axiom 2: conjugate symmetry. Since $H^\dagger = H$ (check immediately from entries), we have

$$\overline{\langle v, u \rangle} = \overline{v^\dagger H u} = u^\dagger H^\dagger v = u^\dagger H v = \langle u, v \rangle.$$

Axiom 3: positive definiteness. Because H is Hermitian, $\langle u, u \rangle = u^\dagger H u \in \mathbb{R}$. We show $u^\dagger H u > 0$ for $u \neq 0$ using Sylvester's criterion (leading principal minors):

$$\det([2]) = 2 > 0, \quad \det \begin{bmatrix} 2 & 1+i \\ 1-i & 3 \end{bmatrix} = 6 - (1+i)(1-i) = 6 - 2 = 4 > 0,$$

and

$$\det(H) = 2 \det \begin{bmatrix} 3 & 2 \\ 2 & 4 \end{bmatrix} - (1+i) \det \begin{bmatrix} 1-i & 2 \\ 0 & 4 \end{bmatrix} = 2(12-4) - (1+i) \cdot 4(1-i) = 16 - 4 \cdot 2 = 8 > 0.$$

Hence H is Hermitian positive definite, so $u^\dagger H u > 0$ for all $u \neq 0$. Therefore $\langle \cdot, \cdot \rangle$ is an inner product.

$$\boxed{\langle \cdot, \cdot \rangle \text{ is an inner product on } \mathbb{C}^{3,1}.}$$

- (b) **Gram–Schmidt on the standard basis.** Let e_1, e_2, e_3 be the standard basis of $\mathbb{C}^{3,1}$. Define

$$u_1 = e_1, \quad u_2 = e_2 - \text{proj}_{u_1}(e_2), \quad u_3 = e_3 - \text{proj}_{u_1}(e_3) - \text{proj}_{u_2}(e_3),$$

$$\text{where } \text{proj}_u(v) = \frac{\langle u, v \rangle}{\langle u, u \rangle} u.$$

Compute the needed inner products using $\langle e_i, e_j \rangle = H_{ij}$:

$$\langle u_1, u_1 \rangle = \langle e_1, e_1 \rangle = 2, \quad \langle u_1, e_2 \rangle = \langle e_1, e_2 \rangle = 1+i, \quad \langle u_1, e_3 \rangle = \langle e_1, e_3 \rangle = 0.$$

Hence

$$u_2 = e_2 - \frac{1+i}{2} e_1.$$

Next,

$$\langle u_2, u_2 \rangle = 2, \quad \langle u_2, e_3 \rangle = 2,$$

so $\text{proj}_{u_2}(e_3) = \frac{2}{2} u_2 = u_2$ and $\text{proj}_{u_1}(e_3) = 0$. Thus

$$u_3 = e_3 - u_2 = e_3 - e_2 + \frac{1+i}{2} e_1.$$

So Gram–Schmidt yields the orthogonal ordered basis

$$\boxed{(u_1, u_2, u_3) = \left(e_1, e_2 - \frac{1+i}{2} e_1, \frac{1+i}{2} e_1 - e_2 + e_3 \right)}.$$

- (c) **Normalize to an orthonormal basis β .** We already computed $\langle u_1, u_1 \rangle = 2$ and $\langle u_2, u_2 \rangle = 2$. Also,

$$\langle u_3, u_3 \rangle = 2$$

(one direct check is $H u_3 = (0, 0, 2)^\top$, hence $u_3^\dagger H u_3 = 2$). Therefore $\|u_k\| = \sqrt{2}$ for $k = 1, 2, 3$.

Define

$$\beta = (\beta_1, \beta_2, \beta_3) := \left(\frac{u_1}{\sqrt{2}}, \frac{u_2}{\sqrt{2}}, \frac{u_3}{\sqrt{2}} \right).$$

Explicitly,

$$\boxed{\beta = \left(\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \frac{1}{\sqrt{2}} \begin{bmatrix} -\frac{1+i}{2} \\ 1 \\ 0 \end{bmatrix}, \frac{1}{\sqrt{2}} \begin{bmatrix} \frac{1+i}{2} \\ -1 \\ 1 \end{bmatrix} \right) \text{ is orthonormal in } (\mathbb{C}^{3,1}, \langle \cdot, \cdot \rangle)}.$$

(d) **β -Fourier coefficients of $v = \begin{bmatrix} 1 \\ 0 \\ i \end{bmatrix}$** . Since β is orthonormal, coefficients are

$$c_k = \langle \beta_k, v \rangle \quad (k = 1, 2, 3).$$

First compute

$$Hv = \begin{bmatrix} 2 & 1+i & 0 \\ 1-i & 3 & 2 \\ 0 & 2 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ i \end{bmatrix} = \begin{bmatrix} 2 \\ 1+i \\ 4i \end{bmatrix}.$$

Now:

$$c_1 = \langle \beta_1, v \rangle = \left(\frac{1}{\sqrt{2}} e_1 \right)^\dagger Hv = \frac{1}{\sqrt{2}} (2) = \sqrt{2}.$$

Let $\alpha := \frac{1+i}{2}$, so $u_2 = e_2 - \alpha e_1$ and $\bar{\alpha} = \frac{1-i}{2}$. Then

$$c_2 = \langle \beta_2, v \rangle = \frac{1}{\sqrt{2}} \langle u_2, v \rangle = \frac{1}{\sqrt{2}} u_2^\dagger (Hv) = \frac{1}{\sqrt{2}} ((-\bar{\alpha}) \cdot 2 + 1 \cdot (1+i)) = \frac{1}{\sqrt{2}} (2i) = \sqrt{2}i.$$

Similarly, $u_3 = \alpha e_1 - e_2 + e_3$, so

$$c_3 = \langle \beta_3, v \rangle = \frac{1}{\sqrt{2}} u_3^\dagger (Hv) = \frac{1}{\sqrt{2}} (\bar{\alpha} \cdot 2 - (1+i) + 4i) = \frac{1}{\sqrt{2}} (2i) = \sqrt{2}i.$$

Therefore the Fourier coefficients are

$$(c_1, c_2, c_3) = (\sqrt{2}, \sqrt{2}i, \sqrt{2}i).$$

Method 2: Matrix/Gram-matrix formulation

The theorems and definitions used in this method are the following.

- **Hermitian positive definite matrix theorem.** If $H \in \mathbb{C}^{n,n}$ is Hermitian positive definite, then

$$\langle u, v \rangle := u^\dagger H v$$

defines an inner product on $\mathbb{C}^{n,1}$ under the convention that the first slot is conjugate-linear and the second slot is linear.

- **Gram matrix definition.** Given a basis $(e_1, \dots, e_n)$ of an inner product space, its Gram matrix is $G = (\langle e_i, e_j \rangle)_{i,j}$. For the standard basis and the inner product $\langle u, v \rangle = u^\dagger H v$, this Gram matrix is exactly H .

(a) Since

$$H^\dagger = H,$$

the matrix H is Hermitian. Its leading principal minors are

$$\Delta_1 = 2 > 0,$$

$$\Delta_2 = \det \begin{bmatrix} 2 & 1+i \\ 1-i & 3 \end{bmatrix} = 6 - (1+i)(1-i) = 6 - 2 = 4 > 0,$$

and

$$\Delta_3 = \det(H) = 8 > 0.$$

By Sylvester's criterion for Hermitian matrices, H is positive definite. Hence, by the Hermitian positive definite matrix theorem, $u^\dagger H v$ defines an inner product.

$\langle \cdot, \cdot \rangle \text{ is an inner product on } \mathbb{C}^{3,1}.$

(b) Let e_1, e_2, e_3 be the standard basis. Because the Gram matrix is H , we can read off

$$\langle e_i, e_j \rangle = H_{ij}.$$

Thus

$$\langle e_1, e_1 \rangle = 2, \quad \langle e_1, e_2 \rangle = 1+i, \quad \langle e_1, e_3 \rangle = 0.$$

Using the projection formula

$$\text{proj}_u(v) = \frac{\langle u, v \rangle}{\langle u, u \rangle} u,$$

we get

$$u_1 = e_1, \quad u_2 = e_2 - \frac{\langle u_1, e_2 \rangle}{\langle u_1, u_1 \rangle} u_1 = e_2 - \frac{1+i}{2} e_1.$$

Next, since

$$\langle u_1, e_3 \rangle = 0, \quad \langle u_2, u_2 \rangle = 2, \quad \langle u_2, e_3 \rangle = 2,$$

we obtain

$$u_3 = e_3 - \frac{\langle u_1, e_3 \rangle}{\langle u_1, u_1 \rangle} u_1 - \frac{\langle u_2, e_3 \rangle}{\langle u_2, u_2 \rangle} u_2 = e_3 - u_2 = \frac{1+i}{2} e_1 - e_2 + e_3.$$

Therefore

$$(u_1, u_2, u_3) = \left(e_1, e_2 - \frac{1+i}{2} e_1, \frac{1+i}{2} e_1 - e_2 + e_3 \right).$$

(c) The same Gram-matrix calculations give

$$\langle u_1, u_1 \rangle = \langle u_2, u_2 \rangle = \langle u_3, u_3 \rangle = 2.$$

Hence

$$\beta = \left(\frac{u_1}{\sqrt{2}}, \frac{u_2}{\sqrt{2}}, \frac{u_3}{\sqrt{2}} \right)$$

is orthonormal, namely

$$\beta = \left(\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \frac{1}{\sqrt{2}} \begin{bmatrix} -\frac{1+i}{2} \\ 1 \\ 0 \end{bmatrix}, \frac{1}{\sqrt{2}} \begin{bmatrix} \frac{1+i}{2} \\ -1 \\ 1 \end{bmatrix} \right).$$

(d) For $v = (1, 0, i)^T$, first compute

$$Hv = \begin{bmatrix} 2 \\ 1+i \\ 4i \end{bmatrix}.$$

Since β is orthonormal, the Fourier coefficients are

$$c_k = \langle \beta_k, v \rangle = \beta_k^\dagger H v.$$

Substituting the three vectors gives

$$c_1 = \sqrt{2}, \quad c_2 = \sqrt{2}i, \quad c_3 = \sqrt{2}i.$$

Therefore

$$(c_1, c_2, c_3) = (\sqrt{2}, \sqrt{2}i, \sqrt{2}i).$$

This method is efficient because it avoids recomputing each inner product from scratch: the entries of H already encode all pairwise inner products of the standard basis vectors.

Method 3: Complete-square / Cholesky factorisation

Let $u = (x, y, z)^\top$. A direct expansion gives

$$u^\dagger H u = 2|x|^2 + (1+i)\bar{x}y + (1-i)\bar{y}x + 3|y|^2 + 2\bar{y}z + 2\bar{z}y + 4|z|^2.$$

This can be written as a sum of squares:

$$u^\dagger H u = 2 \left| x + \frac{1+i}{2}y \right|^2 + 2|y+z|^2 + 2|z|^2.$$

The right-hand side is nonnegative. It is zero only if

$$z = 0, \quad y + z = 0, \quad x + \frac{1+i}{2}y = 0.$$

These equations imply $z = 0$, then $y = 0$, then $x = 0$. Hence

$$u^\dagger H u > 0 \quad (u \neq 0).$$

Since $H^\dagger = H$, the matrix H is Hermitian positive definite. Therefore

$$\langle u, v \rangle = u^\dagger H v$$

defines an inner product.

Equivalently, define

$$R := \begin{bmatrix} 1 & \frac{1+i}{2} & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

Then

$$H = 2R^\dagger R.$$

Thus

$$\langle u, v \rangle = 2(Ru)^\dagger(Rv),$$

so the given inner product is just a positive scalar multiple of the standard inner product after the invertible coordinate transformation $u \mapsto Ru$.

For Gram–Schmidt, use the already computed orthogonal basis

$$u_1 = e_1, \quad u_2 = e_2 - \frac{1+i}{2}e_1, \quad u_3 = e_3 - e_2 + \frac{1+i}{2}e_1.$$

The complete-square factorisation also shows efficiently that

$$\|u_1\|^2 = \|u_2\|^2 = \|u_3\|^2 = 2.$$

Hence the orthonormal basis is

$$\beta = \left(\frac{u_1}{\sqrt{2}}, \frac{u_2}{\sqrt{2}}, \frac{u_3}{\sqrt{2}} \right).$$

For $v = (1, 0, i)^\top$, the Fourier coefficients are again

$$(c_1, c_2, c_3) = (\sqrt{2}, \sqrt{2}i, \sqrt{2}i).$$

Method 4: One-line Gram–Schmidt from the Gram matrix

Because the standard-basis Gram matrix is H , we have

$$\langle e_i, e_j \rangle = H_{ij}.$$

Thus every needed inner product can be read directly from the entries of H .

Seek

$$u_1 = e_1, \quad u_2 = e_2 - ae_1.$$

The condition $u_2 \perp u_1$ gives

$$0 = \langle u_1, u_2 \rangle = \langle e_1, e_2 - ae_1 \rangle = H_{12} - aH_{11}.$$

Therefore

$$a = \frac{H_{12}}{H_{11}} = \frac{1+i}{2},$$

so

$$u_2 = e_2 - \frac{1+i}{2}e_1.$$

Now seek

$$u_3 = e_3 - bu_1 - cu_2.$$

The projection coefficients are

$$b = \frac{\langle u_1, e_3 \rangle}{\langle u_1, u_1 \rangle} = 0, \quad c = \frac{\langle u_2, e_3 \rangle}{\langle u_2, u_2 \rangle} = \frac{2}{2} = 1.$$

Hence

$$u_3 = e_3 - u_2 = e_3 - e_2 + \frac{1+i}{2}e_1.$$

This produces the same answer as Method 1 while minimizing computation:

$$(u_1, u_2, u_3) = \left(e_1, e_2 - \frac{1+i}{2}e_1, e_3 - e_2 + \frac{1+i}{2}e_1 \right).$$

Problem 5 (Double orthogonal complements)

Problem

Let (i) $(V, \langle \cdot, \cdot \rangle)$ be an inner product space, and (ii) W be a vector subspace of V .

- (a) Show that $W \subseteq W^{\perp\perp}$.
- (b) Show that if V is finite-dimensional, then $W = W^{\perp\perp}$.
- (c) Show that if V is infinite-dimensional, then it is not necessarily true that $W = W^{\perp\perp}$.

Solution

Method 1: Direct definition + dimension counting (finite-dimensional case)

- (a) Let $w \in W$. To show $w \in W^{\perp\perp}$ we must show $\langle w, u \rangle = 0$ for all $u \in W^\perp$. But $u \in W^\perp$ means precisely that $\langle u, w' \rangle = 0$ for all $w' \in W$; in particular $\langle u, w \rangle = 0$. Using conjugate symmetry (or symmetry in the real case), $\langle w, u \rangle = 0$. Hence $w \in W^{\perp\perp}$. So $W \subseteq W^{\perp\perp}$.

$$\boxed{W \subseteq W^{\perp\perp}}$$

- (b) Assume $\dim(V) < \infty$. A standard theorem gives

$$\dim(W) + \dim(W^\perp) = \dim(V).$$

Apply this to $W^\perp$:

$$\dim(W^\perp) + \dim(W^{\perp\perp}) = \dim(V).$$

Subtracting yields $\dim(W^{\perp\perp}) = \dim(W)$. Together with (a) and finite-dimensionality,

$$W \subseteq W^{\perp\perp} \text{ and } \dim(W) = \dim(W^{\perp\perp}) \implies W = W^{\perp\perp}.$$

$$\boxed{\dim(V) < \infty \implies W = W^{\perp\perp}}$$

- (c) In infinite dimensions, equality can fail. Let $V = \ell^2$ be the space of square-summable real sequences (with the standard inner product $\langle x, y \rangle = \sum_{k=1}^{\infty} x_k y_k$), and let

$$W = c_{00} := \{x \in \ell^2 : x \text{ has finite support}\}.$$

Then W is a subspace of V and is infinite-dimensional (but not all of ℓ^2).

We claim $W^\perp = \{0\}$. Indeed, if $z \in W^\perp$, then for each standard basis vector $e_k \in c_{00}$ we have

$$0 = \langle z, e_k \rangle = z_k,$$

so all coordinates of z are 0, hence $z = 0$. Therefore $W^\perp = \{0\}$ and hence $W^{\perp\perp} = V$.

But $W \neq V$ (e.g. $(1, 1/2, 1/3, \dots) \in \ell^2$ but not finitely supported). Thus $W \neq W^{\perp\perp}$.

$$\boxed{\text{In infinite dimensions, } W \neq W^{\perp\perp} \text{ can occur (e.g. } W = c_{00} \subset \ell^2\text{).}}$$

Method 2: Orthogonal decomposition (finite-dimensional case) + “closure” intuition

- (a) Same as Method 1.
- (b) If $\dim(V) < \infty$, choose an orthonormal basis of W and extend it to an orthonormal basis of V by Gram–Schmidt:

$$(\text{ONB of } W) \cup (\text{ONB of } W^\perp) = \text{ONB of } V.$$

Then every $v \in V$ has a unique decomposition $v = w + u$ with $w \in W$, $u \in W^\perp$. Now if $v \in W^{\perp\perp}$, then $\langle v, u \rangle = 0$ for all $u \in W^\perp$. In particular, taking u from the decomposition $v = w + u$ gives

$$0 = \langle v, u \rangle = \langle w + u, u \rangle = \langle w, u \rangle + \langle u, u \rangle = 0 + \|u\|^2,$$

so $u = 0$ and hence $v = w \in W$. Thus $W^{\perp\perp} \subseteq W$, and with (a) we get equality.

- (c) In general (especially in infinite dimensions), $W^{\perp\perp}$ corresponds to the “closed span” of W in the norm topology, so if W is not closed, strict inclusion may occur. The ℓ^2 example above realizes this phenomenon concretely.

Method 3: Projection theorem proof in finite dimensions

The key finite-dimensional theorem is the orthogonal decomposition theorem:

$$V = W \oplus W^\perp.$$

Thus every $v \in V$ has a unique decomposition

$$v = w + u, \quad w \in W, \quad u \in W^\perp.$$

We already know from Part (a) that

$$W \subseteq W^{\perp\perp}.$$

To prove the reverse inclusion in finite dimensions, let $v \in W^{\perp\perp}$. Write

$$v = w + u, \quad w \in W, \quad u \in W^\perp.$$

Since $v \in W^{\perp\perp}$, it is orthogonal to every vector in $W^\perp$, in particular to u . Therefore

$$0 = \langle v, u \rangle = \langle w + u, u \rangle = \langle w, u \rangle + \langle u, u \rangle.$$

But $w \in W$ and $u \in W^\perp$, so $\langle w, u \rangle = 0$. Hence

$$\|u\|^2 = 0,$$

so $u = 0$. Therefore

$$v = w \in W.$$

Thus

$$W^{\perp\perp} \subseteq W.$$

Together with $W \subseteq W^{\perp\perp}$, this gives

$$\boxed{W = W^{\perp\perp} \quad \text{if } \dim V < \infty.}$$

This proof is often more conceptual than dimension counting because it explicitly uses the perpendicular component of v .

Method 4: Closure theorem in Hilbert spaces

In a Hilbert space, the stronger theorem is

$$\boxed{W^{\perp\perp} = \overline{W}.}$$

Here $\overline{W}$ is the closure of W under the norm induced by the inner product.

This theorem explains both the finite-dimensional and infinite-dimensional cases at once. In finite dimensions, every subspace is closed, so

$$\overline{W} = W,$$

and hence

$$W^{\perp\perp} = W.$$

In infinite dimensions, a subspace need not be closed. If W is dense but not equal to V , then

$$\overline{W} = V,$$

so

$$W^{\perp\perp} = V \neq W.$$

For the example in Method 1,

$$W = c_{00} \subset \ell^2.$$

The subspace c_{00} of finitely supported sequences is dense in ℓ^2 , so

$$\overline{c_{00}} = \ell^2.$$

Therefore

$$c_{00}^{\perp\perp} = \ell^2 \neq c_{00}.$$

This is the advanced conceptual reason why Part (c) is possible.

Problem 6 (Orthogonal complement and orthogonal decomposition in $\mathbb{R}^4$)

Problem

Let

$$V := \{(w, x, y, z) \in \mathbb{R}^4 \mid w - y + z = 0\},$$

which is a vector subspace of $\mathbb{R}^4$ (standard inner product $\cdot$).

- (a) Derive a basis for $V^\perp$.
- (b) Derive an orthonormal basis for $(V, \cdot)$.
- (c) Compute the unique $a \in V$ and $b \in V^\perp$ such that $(1, 2, 3, 4) = a + b$.

Solution

Method 1: Solve linear constraints for $V^\perp$; then Gram–Schmidt inside V

- (a) Parameterize V by free variables x, y, z :

$$w = y - z, \quad (w, x, y, z) = (y - z, x, y, z) = x(0, 1, 0, 0) + y(1, 0, 1, 0) + z(-1, 0, 0, 1).$$

Let $u = (a, b, c, d) \in \mathbb{R}^4$. Then $u \in V^\perp$ iff $u \cdot v = 0$ for all $v \in V$. It suffices to impose orthogonality to the spanning vectors:

$$u \cdot (0, 1, 0, 0) = b = 0, \quad u \cdot (1, 0, 1, 0) = a + c = 0, \quad u \cdot (-1, 0, 0, 1) = -a + d = 0.$$

Thus $b = 0$, $c = -a$, $d = a$, so

$$V^\perp = \text{span}\{(1, 0, -1, 1)\}.$$

$$\boxed{V^\perp = \text{span}\{(1, 0, -1, 1)\}}.$$

- (b) A convenient basis of V from the parameterization is

$$v_1 = (0, 1, 0, 0), \quad v_2 = (1, 0, 1, 0), \quad v_3 = (-1, 0, 0, 1).$$

Apply Gram–Schmidt:

Step 1: $e_1 = v_1$ (already unit), so $e_1 = (0, 1, 0, 0)$.

Step 2: $v_2 \cdot e_1 = 0$, so $u_2 = v_2$ and

$$e_2 = \frac{v_2}{\|v_2\|} = \frac{1}{\sqrt{2}}(1, 0, 1, 0).$$

Step 3: Orthogonalize v_3 against $u_2 = v_2$:

$$v_3 \cdot v_2 = (-1) \cdot 1 + 0 \cdot 0 + 0 \cdot 1 + 1 \cdot 0 = -1, \quad v_2 \cdot v_2 = 2.$$

So

$$u_3 = v_3 - \text{proj}_{v_2}(v_3) = v_3 - \frac{-1}{2}v_2 = v_3 + \frac{1}{2}v_2 = \left(-\frac{1}{2}, 0, \frac{1}{2}, 1\right).$$

Equivalently, $2u_3 = (-1, 0, 1, 2)$, and $\|(-1, 0, 1, 2)\|^2 = 6$. Hence

$$e_3 = \frac{1}{\sqrt{6}}(-1, 0, 1, 2).$$

Thus an orthonormal basis for V is

$$\left((0, 1, 0, 0), \frac{1}{\sqrt{2}}(1, 0, 1, 0), \frac{1}{\sqrt{6}}(-1, 0, 1, 2) \right).$$

(c) Let $p = (1, 2, 3, 4)$. Since $V^\perp = \text{span}\{u\}$ with $u = (1, 0, -1, 1)$, we have

$$b = \text{proj}_{V^\perp}(p) = \frac{p \cdot u}{u \cdot u}u.$$

Compute

$$p \cdot u = 1 \cdot 1 + 2 \cdot 0 + 3 \cdot (-1) + 4 \cdot 1 = 2, \quad u \cdot u = 1 + 0 + 1 + 1 = 3.$$

So

$$b = \frac{2}{3}(1, 0, -1, 1) = \left(\frac{2}{3}, 0, -\frac{2}{3}, \frac{2}{3}\right).$$

Then

$$a = p - b = \left(1 - \frac{2}{3}, 2, 3 + \frac{2}{3}, 4 - \frac{2}{3}\right) = \left(\frac{1}{3}, 2, \frac{11}{3}, \frac{10}{3}\right).$$

Check $a \in V$:

$$w - y + z = \frac{1}{3} - \frac{11}{3} + \frac{10}{3} = 0.$$

Therefore

$$a = \left(\frac{1}{3}, 2, \frac{11}{3}, \frac{10}{3}\right) \in V, \quad b = \left(\frac{2}{3}, 0, -\frac{2}{3}, \frac{2}{3}\right) \in V^\perp, \quad (1, 2, 3, 4) = a + b.$$

Method 2: Matrix viewpoint (constraint normal vector)

The subspace $V = \{x \in \mathbb{R}^4 : n \cdot x = 0\}$ where $n = (1, 0, -1, 1)$ (since $w - y + z = 0$). Hence $V^\perp = \text{span}\{n\}$ immediately. The orthogonal decomposition of p is

$$p = \left(p - \frac{p \cdot n}{n \cdot n}n\right) + \frac{p \cdot n}{n \cdot n}n,$$

matching the a, b above.

Method 3: Kernel–row-space theorem

Write the defining equation of V as a matrix equation:

$$w - y + z = 0 \iff \begin{bmatrix} 1 & 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} w \\ x \\ y \\ z \end{bmatrix} = 0.$$

Hence

$$V = \text{Null}(A), \quad A := \begin{bmatrix} 1 & 0 & -1 & 1 \end{bmatrix}.$$

A standard theorem says

$$\text{Null}(A)^\perp = \text{Row}(A).$$

Therefore

$$V^\perp = \text{Row}(A) = \text{span}\{(1, 0, -1, 1)\}.$$

So

$$\boxed{V^\perp = \text{span}\{(1, 0, -1, 1)\}.$$

The orthogonal decomposition of $p = (1, 2, 3, 4)$ then follows by projecting onto the row vector

$$n = (1, 0, -1, 1).$$

Thus

$$b = \frac{p \cdot n}{n \cdot n} n = \frac{2}{3} (1, 0, -1, 1) = \left(\frac{2}{3}, 0, -\frac{2}{3}, \frac{2}{3} \right),$$

and

$$a = p - b = \left(\frac{1}{3}, 2, \frac{11}{3}, \frac{10}{3} \right).$$

Method 4: Lagrange multiplier / closest-point method

The vector $a \in V$ in the decomposition

$$p = a + b, \quad a \in V, \quad b \in V^\perp,$$

is the closest point in the hyperplane V to $p = (1, 2, 3, 4)$. Thus we minimize

$$\|p - a\|^2$$

subject to the constraint

$$n \cdot a = 0, \quad n = (1, 0, -1, 1).$$

The closest-point condition says that $p - a$ must be normal to the hyperplane, so

$$p - a = \lambda n.$$

Hence

$$a = p - \lambda n.$$

Impose $a \in V$:

$$0 = n \cdot a = n \cdot (p - \lambda n) = n \cdot p - \lambda n \cdot n.$$

Therefore

$$\lambda = \frac{n \cdot p}{n \cdot n} = \frac{2}{3}.$$

So

$$b = p - a = \lambda n = \frac{2}{3}(1, 0, -1, 1),$$

and

$$a = p - b = \left(\frac{1}{3}, 2, \frac{11}{3}, \frac{10}{3} \right).$$

This method is especially useful for projection questions involving one linear constraint.

Method 5: Constructing the orthonormal basis by inspection

Since

$$V = n^\perp, \quad n = (1, 0, -1, 1),$$

we need three mutually orthogonal unit vectors perpendicular to n . One immediate choice is

$$e_1 = (0, 1, 0, 0),$$

because its dot product with n is 0. Now look only at the (w, y, z) coordinates satisfying

$$w - y + z = 0.$$

Two convenient perpendicular choices are

$$v_2 = (1, 0, 1, 0), \quad v_3 = (-1, 0, 1, 2).$$

Indeed,

$$v_2 \cdot n = 1 - 1 = 0, \quad v_3 \cdot n = -1 - 1 + 2 = 0,$$

and

$$v_2 \cdot v_3 = -1 + 1 = 0.$$

Also both are perpendicular to e_1 . Their norms are

$$\|e_1\| = 1, \quad \|v_2\| = \sqrt{2}, \quad \|v_3\| = \sqrt{6}.$$

Thus an orthonormal basis is

$$\left((0, 1, 0, 0), \frac{1}{\sqrt{2}}(1, 0, 1, 0), \frac{1}{\sqrt{6}}(-1, 0, 1, 2) \right).$$

This gives the same basis as Method 1 but avoids running the full Gram–Schmidt algorithm.

Problem 7 (Diagonal matrices and Frobenius orthogonal complement)

Problem

Let $n \in \mathbb{N}$, and consider the real inner product space $(\mathbb{R}^{n,n}, \langle \cdot, \cdot \rangle_F)$, where

$$\langle A, B \rangle_F := \text{trace}(A^T B) = \sum_{i=1}^n \sum_{j=1}^n a_{ij} b_{ij}.$$

Let

$$D := \{A \in \mathbb{R}^{n,n} \mid A \text{ is diagonal}\}.$$

Derive a basis for $D^\perp$.

Solution

Method 1: Characterize $D^\perp$ by testing against diagonal matrices

Let $X = (x_{ij}) \in \mathbb{R}^{n,n}$. Then $X \in D^\perp$ iff $\langle X, \Delta \rangle_F = 0$ for all diagonal $\Delta = \text{diag}(d_1, \dots, d_n)$. But

$$\langle X, \Delta \rangle_F = \sum_{i=1}^n \sum_{j=1}^n x_{ij} \Delta_{ij} = \sum_{i=1}^n x_{ii} d_i.$$

For this to be 0 for all choices of $(d_1, \dots, d_n) \in \mathbb{R}^n$, we must have $x_{ii} = 0$ for every i . Thus

$$D^\perp = \{X \in \mathbb{R}^{n,n} : x_{11} = x_{22} = \dots = x_{nn} = 0\},$$

i.e. the subspace of matrices with zero diagonal.

Let E_{ij} be the standard matrix units. Then $\{E_{ij} : i \neq j\}$ spans precisely the zero-diagonal matrices and is linearly independent. Therefore

$$D^\perp = \text{span}\{E_{ij} : i \neq j\} \quad \text{and} \quad \{E_{ij} : i \neq j\} \text{ is a basis of } D^\perp.$$

Method 2: Orthogonal direct sum decomposition

Every matrix $A \in \mathbb{R}^{n,n}$ decomposes uniquely as

$$A = \underbrace{\text{diag}(a_{11}, \dots, a_{nn})}_{\in D} + \underbrace{(A - \text{diag}(a_{11}, \dots, a_{nn}))}_{\text{zero diagonal}}.$$

Moreover, diagonal matrices are Frobenius-orthogonal to zero-diagonal matrices, since

$$\langle \text{diag}(d_1, \dots, d_n), X \rangle_F = \sum_{i=1}^n d_i x_{ii} = 0$$

whenever $x_{ii} = 0$. Hence $\mathbb{R}^{n,n} = D \oplus D^\perp$ and $D^\perp$ is exactly the zero-diagonal subspace, with basis $\{E_{ij} : i \neq j\}$.

Method 3: Matrix-unit orthonormal basis

Let E_{ij} denote the standard matrix unit whose (i, j) -entry is 1 and whose other entries are 0. Under the Frobenius inner product,

$$\langle E_{ij}, E_{kl} \rangle_F = \text{trace}(E_{ij}^\top E_{kl}) = \delta_{ik} \delta_{jl}.$$

Therefore

$$\{E_{ij} : 1 \leq i, j \leq n\}$$

is an orthonormal basis of $\mathbb{R}^{n,n}$. The diagonal subspace is

$$D = \text{span}\{E_{11}, E_{22}, \dots, E_{nn}\}.$$

Since the full matrix-unit basis is orthonormal, the orthogonal complement of D is spanned by all matrix units not appearing in the basis for D :

$$D^\perp = \text{span}\{E_{ij} : i \neq j\}.$$

This is the fastest method because it identifies D as a coordinate subspace of the Frobenius matrix space.

Method 4: Orthogonal projection onto the diagonal subspace

Define the diagonal projection

$$\Pi_D(A) := \text{diag}(a_{11}, a_{22}, \dots, a_{nn}).$$

Then every matrix $A \in \mathbb{R}^{n,n}$ decomposes uniquely as

$$A = \Pi_D(A) + (A - \Pi_D(A)).$$

The first term is diagonal, and the second term has zero diagonal. Moreover, if $\Delta = \text{diag}(d_1, \dots, d_n)$ and X has zero diagonal, then

$$\langle \Delta, X \rangle_F = \sum_{i=1}^n d_i x_{ii} = 0.$$

Hence the decomposition is an orthogonal direct sum:

$$\mathbb{R}^{n,n} = D \oplus D^\perp.$$

Therefore

$$D^\perp = \ker(\Pi_D) = \{X \in \mathbb{R}^{n,n} : x_{11} = x_{22} = \dots = x_{nn} = 0\}.$$

A basis is again

$$\{E_{ij} : i \neq j\}.$$